TOR AMPLITUDE IS THE SAME OVER A LOCAL RING, OVER ITS HENSELIZATION AND OVER ITS STRICT HENSELIZATION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file flat.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the chapter's own retained minimal preamble (sources/stacks-flat/retained-preamble.tex, lines 1-85) is a verbatim excerpt of the native Xy-pic setup and of the native theorem declarations and mathematical macros, followed by the project's own reference/slogan/history display-environment definitions copied mechanically from the shared preamble; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler, and extra_packages is left empty because everything needed is already in the retained preamble. No macro was invented or redefined.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- Henselization and strict henselization, and faithful flatness of the local ring maps. Algebra, label algebra-definition-henselization (statement, proof line 248); More on Algebra, label more-algebra-lemma-dumb-properties-henselization (proof line 274). Fixes the rings considered and gives that the local ring maps from the local ring to its henselization and strict henselization are faithfully flat, which yields the easy implications by pushforward of tor amplitude.
- Pushforward and descent of tor amplitude along faithfully flat ring maps. More on Algebra, labels more-algebra-lemma-flat-push-tor-amplitude (proof line 280) and more-algebra-lemma-flat-descent-tor-amplitude (proof line 283). The first gives the implications from the strict henselization down to the henselization and to the local ring, and the second gives the equivalences between the cases (6), (5) and (3) by descent.
- Left resolutions by free modules, flatness of a last cokernel and the flat up and down lemma. Derived Categories, label derived-lemma-subcategory-left-resolution (proof line 293); More on Algebra, labels more-algebra-lemma-last-one-flat (proof line 296) and more-algebra-lemma-tor-amplitude (proof line 304); Same chapter flat.tex, label lemma-flat-up-down-henselization (proof line 298). Produces the bounded above complex of free modules representing the object, identifies a cokernel of its differentials as a flat module over the local ring, transports flatness to the strict henselization, and thereby exhibits a complex of flat modules representing the object, which is tor amplitude.

Date: stacks-project:flat.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original definition 04GQ: sources/stacks-chow/tags.txt, source lines 5841–5841. Navigation pointer for \ref{algebra-definition-henselization} used inside the counted span (line 248); the argument that consumes it is retained in the span.

Source reference. Original lemma 05T7: sources/stacks-chow/tags.txt, source lines 7471–7471. Navigation pointer for \ref{derived-lemma-subcategory-left-resolution} used inside the counted span (line 293); the argument that consumes it is retained in the span.

Source reference. Original lemma 05VL: sources/stacks-flat/pinned-original-source.tex, source lines 183–183. Navigation pointer for \ref{lemma-flat-up-down-henselization} used inside the counted span (line 298); the argument that consumes it is retained in the span.

Source reference. Original lemma 07QM: sources/stacks-chow/tags.txt, source lines 9837–9837. Navigation pointer for \ref{more-algebra-lemma-dumb-properties-henselization} used inside the counted span (line 274); the argument that consumes it is retained in the span.

Source reference. Original lemma 068S: sources/stacks-chow/tags.txt, source lines 8016–8016. Navigation pointer for \ref{more-algebra-lemma-flat-descent-tor-amplitude} used inside the counted span (line 283); the argument that consumes it is retained in the span.

Source reference. Original lemma 066J: sources/stacks-chow/tags.txt, source lines 7938–7938. Navigation pointer for \ref{more-algebra-lemma-flat-push-tor-amplitude} used inside the counted span (line 280); the argument that consumes it is retained in the span.

Source reference. Original lemma 0653: sources/stacks-chow/tags.txt, source lines 7887–7887. Navigation pointer for \ref{more-algebra-lemma-last-one-flat} used inside the counted span (line 296); the argument that consumes it is retained in the span.

Source reference. Original lemma 0654: sources/stacks-chow/tags.txt, source lines 7888–7888. Navigation pointer for \ref{more-algebra-lemma-tor-amplitude} used inside the counted span (line 304); the argument that consumes it is retained in the span.

Lemma 0.1. *Let S be a scheme and $s \in S$ a point. Denote $\mathcal{O}^h_{S,s}$ (resp. $\mathcal{O}^{sh}_{S,s}$) the henselization (resp. strict henselization), see Algebra, Definition 04GQ. Let M^{sh} be an object of $D(\mathcal{O}^{sh}_{S,s})$. Let $a, b \in \mathbf{Z}$. The following are equivalent*

(1) *M^{sh} has tor amplitude in $[a, b]$ over $\mathcal{O}_{S,s}$,*
(2) *M^{sh} has tor amplitude in $[a, b]$ over $\mathcal{O}^h_{S,s}$, and*
(3) *M^{sh} has tor amplitude in $[a, b]$ over $\mathcal{O}^{sh}_{S,s}$.*

If $M^{sh} = M^h \otimes^{\mathbf{L}}_{\mathcal{O}^h_{S,s}} \mathcal{O}^{sh}_{S,s}$ for $M^h \in D(\mathcal{O}^h_{S,s})$ this is also equivalent to

(4) *M^h has tor amplitude in $[a, b]$ over $\mathcal{O}_{S,s}$, and*
(5) *M^h has tor amplitude in $[a, b]$ over $\mathcal{O}^h_{S,s}$.*

If $M^h = M \otimes^{\mathbf{L}}_{\mathcal{O}_{S,s}} \mathcal{O}^h_{S,s}$ for $M \in D(\mathcal{O}_{S,s})$ this is also equivalent to

(6) *M has tor amplitude in $[a, b]$ over $\mathcal{O}_{S,s}$.*

Proof. By More on Algebra, Lemma 07QM the local ring maps $\mathcal{O}_{S,s} \to \mathcal{O}^h_{S,s} \to \mathcal{O}^{sh}_{S,s}$ are faithfully flat. Hence (3) $\Rightarrow$ (2) $\Rightarrow$ (1) and (5) $\Rightarrow$ (4) follow from More on Algebra, Lemma 066J. By faithful flatness the equivalences (6) $\Leftrightarrow$ (5) and (5) $\Leftrightarrow$

(3) follow from More on Algebra, Lemma 068S. Thus it suffices to show that (1) $\Rightarrow$ (3), (2) $\Rightarrow$ (3), and (4) $\Rightarrow$ (5).

Assume (1). In particular M^{sh} has vanishing cohomology in degrees $< a$ and $> b$. Hence we can represent M^{sh} by a complex $P^\bullet$ of free $\mathcal{O}_{X,x}^{sh}$-modules with $P^i = 0$ for $i > b$ (see for example the very general Derived Categories, Lemma 05T7). Note that P^n is flat over $\mathcal{O}_{S,s}$ for all n. Consider $\mathrm{Coker}(d_P^{a-1})$. By More on Algebra, Lemma 0653 this is a flat $\mathcal{O}_{S,s}$-module. Hence by Lemma 05VL this is a flat $\mathcal{O}_{S,s}^{sh}$-module. Thus $\tau_{\geq a}P^\bullet$ is a complex of flat $\mathcal{O}_{S,s}^{sh}$-modules representing M^{sh} in $D(\mathcal{O}_{S,s}^{sh}$ and we find that M^{sh} has tor amplitude in $[a, b]$, see More on Algebra, Lemma 0654. Thus (1) implies (3). Of course this implies also (2) $\Rightarrow$ (3) by replacing $\mathcal{O}_{S,s}$ by $\mathcal{O}_{S,s}^h$. The same argument applies to prove (4) $\Rightarrow$ (5). $\square$